

Supplementary Information

When Better QAOA Schedules Depend on the Simulator: Truncation-Induced Rank Reversals on a 55-Qubit QOBLIB Benchmark

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1 Frozen protocol and evaluated schedules

The split and primary promotion gate were written before any candidate was evaluated on `es60fst02`. Training used the complete `es60fst01` and `es60fst03` graphs, validation used `es60fst04`, and `es60fst02` was held out for one confirmatory pass. The gate required a positive paired BKS-rate difference from published LR, a paired bootstrap 95% interval excluding zero, and no material loss of feasible mass. The nonlinear schedule family was

$$\beta_k = \Delta_\beta \left(\frac{p-k+1}{p} \right)^{a_\beta}, \quad \gamma_k = \Delta_\gamma \left(\frac{k}{p} \right)^{a_\gamma}, \quad p = 15.$$

Table 1: Schedules frozen before the blind pass. The genome columns are $(\Delta_\beta, \Delta_\gamma, a_\beta, a_\gamma)$.

Method	Δ_β	Δ_γ	a_β	a_γ
Published LR	0.70000	0.40000	1.00000	1.00000
Evolutionary	0.51750	0.77197	1.07734	1.75435
Matched random	0.64247	0.75939	1.77679	0.99172

Three evolutionary replicates used a fixed population of 20 for six generations. Uniform random search received exactly 120 independent candidates per replicate. Every candidate received 256 exact-statevector shots on each training kernel. One champion per search method was selected on validation; test metrics never influenced selection. Samples were only unfolded and checked for independence on the original graph. No repair, greedy fill, local search, or archived solution was used.

The MIS QUBO penalty was $\lambda = 1.5$, with the released maximum quadratic-coefficient normalization. The 186-vertex, 280-edge blind graph was reduced to a 55-qubit, 91-edge kernel. Its native depth-15 circuit contains 1,365 RZZ, 825 RZ, 825 RX, and 55 Hadamard operations before measurement.

2 Blind paired analysis

The confirmatory experiment used 15 paired simulator seeds and 1,000 shots per method and seed. Table 2 reports pooled rates; Table 3 reports paired inference. The confidence interval resamples the 15 paired jobs with 20,000 bootstrap draws. The two-sided sign-flip test enumerates all 2^{15} paired sign assignments. This preserves job-level pairing rather than treating 15,000 individual shots as independent simulator replicates.

Table 2: Blind `es60fst02` results at bond 64 and $\tau_{\text{MPS}} = 10^{-3}$. Each method has 15,000 shots.

Method	BKS hits	BKS rate	BKS-1	Feasible	Quality	N_{95}
Published LR	41	.00273	.11213	.50127	.48962	1,095
Evolutionary	31	.00207	.06813	.89987	.87202	1,449
Matched random	101	.00673	.15700	.90180	.87865	444

Here $N_{95} = \lceil \log(0.05) / \log(1 - p_{\text{BKS}}) \rceil$ is the estimated number of shots required for a 95% chance of at least one BKS sample.

Table 3: Paired method-minus-LR comparisons across 15 jobs. W/T/L denotes wins, ties, and losses.

Method	Metric	Mean Δ	Bootstrap 95% CI	p_{flip}	W/T/L
Evolutionary	BKS	−.00067	[−.00167, .00033]	.27002	5/2/8
Evolutionary	Feasible	+.39860	[.39133, .40660]	6.10×10^{-5}	15/0/0
Matched random	BKS	+.00400	[.00247, .00553]	.000244	13/2/0
Matched random	Near-BKS	+.04487	[.03780, .05140]	6.10×10^{-5}	15/0/0
Matched random	Feasible	+.40053	[.39266, .40987]	6.10×10^{-5}	15/0/0
Matched random	Quality	+.38904	[.38140, .39807]	6.10×10^{-5}	15/0/0

3 Complete 55-qubit sensitivity audit

All sensitivity points are single jobs with 10,000 shots per method, except the primary setting, which pools the 15 confirmatory jobs. Wilson intervals are stored with the artifact. Table 4 exposes the integer hit counts so every displayed rate can be reconstructed.

Table 4: All completed native-circuit MPS settings on the held-out 55-qubit kernel. RS is matched random search.

Bond	τ_{MPS}	Method	Shots	BKS	BKS−1	Feasible	Quality
64	10^{-3}	LR	15,000	41 (.0027)	1,682 (.1121)	7,519 (.5013)	.4896
64	10^{-3}	RS	15,000	101 (.0067)	2,355 (.1570)	13,527 (.9018)	.8787
64	3×10^{-4}	LR	10,000	116 (.0116)	1,542 (.1542)	4,805 (.4805)	.4702
64	3×10^{-4}	RS	10,000	161 (.0161)	1,949 (.1949)	8,129 (.8129)	.7934
64	10^{-4}	LR	10,000	153 (.0153)	1,546 (.1546)	4,809 (.4809)	.4706
64	10^{-4}	RS	10,000	109 (.0109)	1,621 (.1621)	7,351 (.7351)	.7170
96	10^{-3}	LR	10,000	19 (.0019)	1,188 (.1188)	5,055 (.5055)	.4938
96	10^{-3}	RS	10,000	70 (.0070)	1,661 (.1661)	8,925 (.8925)	.8701
96	10^{-4}	LR	10,000	152 (.0152)	1,710 (.1710)	4,799 (.4799)	.4700
96	10^{-4}	RS	10,000	134 (.0134)	1,808 (.1808)	7,121 (.7121)	.6953

The BKS ranking favors RS at 10^{-3} and 3×10^{-4} , but LR at 10^{-4} for both bond caps. Near-BKS, feasible, and quality mass favor RS at all completed points. Raising the bond cap from 64 to 96 does not remove the rank reversal. The attempted bond-128, 10^{-6} , three-method batch exceeded 20 minutes and was terminated without producing a result; it is not treated as a completed point.

4 Exact-state MPS calibration

Complete output states were computed for the real 15-qubit **es60fst01** and 12-qubit **es60fst03** kernels. Probabilities were compared before measurement, eliminating shot noise. For exact distribution p and MPS distribution q , total variation is $\frac{1}{2} \sum_x |p(x) - q(x)|$; fidelity is the squared state overlap. Table 5 gives all bond-128 threshold points.

At each of 10^{-3} , 3×10^{-4} , and 10^{-4} , the state and metric errors are numerically identical at bonds 32, 64, and 128. Bond 16 is also identical except for nonlinear **es60fst01** at 10^{-4} : TV is 0.032981 rather than 0.032493 and BKS error is 0.013332 rather than 0.012953. Thus the threshold dominates once the cap reaches 32 on these donor kernels. The signed LR BKS error crosses zero twice even while fidelity improves, so a monotone global state metric does not certify monotone convergence of a rare observable.

5 Independent cuTensorNet audit

We exported the same native 55-qubit circuits to NVIDIA cuTensorNet 26.6 and decoded samples with the unchanged QOBLIB reduction map. An independent exact contraction on the 12- and 15-qubit donor kernels agreed with Qiskit statevectors to unit fidelity for both schedules and both qubit orderings. On the 55-qubit kernel, the exact sampler failed during contraction planning with an internal backend error. We therefore report the successful cuTensorNet MPS sweep, including the failed attempt in the audit log.

A Fiedler ordering of the interaction graph reduced bandwidth from 47 to 10 and maximum cut width from 42 to 9. It is an exact relabeling: samples were inverse-permuted before decoding. This reduced the bond-32, 100-shot runtime

Table 5: Exact-state calibration. p_{BKS}^* is exact BKS probability and $\Delta p_{\text{BKS}} = p_{\text{BKS}}^{\text{MPS}} - p_{\text{BKS}}^*$. NL is the matched-random nonlinear schedule.

Case	Method	τ_{MPS}	Fidelity	TV	p_{BKS}^*	Δp_{BKS}
01	LR	10^{-3}	.980259	.051752	.686100	+.015769
01	LR	3×10^{-4}	.991631	.036703	.686100	+.009153
01	LR	10^{-4}	.995614	.025605	.686100	-.001314
01	LR	10^{-5}	.998979	.008277	.686100	-.000559
01	LR	10^{-6}	.999866	.002659	.686100	+.000153
01	NL	10^{-3}	.952413	.074245	.928984	+.034173
01	NL	3×10^{-4}	.974923	.052601	.928984	+.022460
01	NL	10^{-4}	.986271	.032493	.928984	+.012953
01	NL	10^{-5}	.998091	.008755	.928984	+.002279
01	NL	10^{-6}	.999836	.001448	.928984	+.000220
03	LR	10^{-3}	.987431	.050922	.735826	+.007444
03	LR	3×10^{-4}	.994573	.021938	.735826	+.003234
03	LR	10^{-4}	.997701	.016650	.735826	-.002770
03	LR	10^{-5}	.999535	.004628	.735826	-.000237
03	LR	10^{-6}	.999953	.001805	.735826	+.000204
03	NL	10^{-3}	.976398	.070154	.955300	+.018487
03	NL	3×10^{-4}	.986230	.050596	.955300	+.010784
03	NL	10^{-4}	.994029	.024675	.955300	+.005526
03	NL	10^{-5}	.999340	.005522	.955300	+.000701
03	NL	10^{-6}	.999969	.001053	.955300	+.000133

from 428–463 seconds per schedule in sorted order to 65–69 seconds. Table 6 gives the completed independent sweep. Near means objective at least BKS–1.

Table 6: Independent cuTensorNet MPS results on `es60fst02`. NL is the frozen matched-random nonlinear schedule.

Order	Bond	τ_{MPS}	Method	Shots	BKS	Near	Feasible	Seconds
Sorted	32	10^{-3}	LR	100	0	.000	.250	428.3
Sorted	32	10^{-3}	NL	100	0	.000	.700	463.3
Fiedler	32	10^{-3}	LR	100	0	.030	.440	64.7
Fiedler	32	10^{-3}	NL	100	0	.040	.740	68.7
Fiedler	64	10^{-3}	LR	200	1 (.0050)	.020	.440	113.6
Fiedler	64	10^{-3}	NL	200	1 (.0050)	.065	.725	117.0
Fiedler	128	10^{-3}	LR	5,000	6 (.0012)	.0424	.4840	228.7
Fiedler	128	10^{-3}	NL	5,000	15 (.0030)	.0820	.6232	252.7
Fiedler	128	10^{-4}	LR	5,000	12 (.0024)	.0764	.4838	222.2
Fiedler	128	10^{-4}	NL	5,000	11 (.0022)	.0782	.6352	260.1

At 10^{-3} , NL has 15 versus 6 BKS hits (Fisher two-sided $p = .078$); at 10^{-4} the counts are 11 versus 12 ($p = 1.0$). Thus the independent backend reproduces loss of the apparent NL advantage under tighter truncation, but not a statistically resolved reversal. Feasible and near-BKS mass favor NL at both thresholds.

6 Strong classical MIS controls

The complete 186-vertex graph was also solved directly with two classical controls. SciPy/HiGHS mixed-integer programming certified the optimum 88 with zero reported gap in 0.036 seconds. The stochastic control repeatedly picks a minimum residual-degree vertex with random tie breaking. It uses no archived solution, parameter training, or repair after sampling. We ran 15 independent replicates of 1,000 starts on both the full graph and the released 55-variable reduced kernel.

Full-graph and kernel greedy achieve 11.2 and 60.0 times the QAOA BKS rate, respectively, while also running faster. The exact solver is faster still on this instance. These controls rule out a competitive or quantum-advantage claim for the present experiment; the contribution is the cross-backend diagnosis of rare-event instability.

Table 7: Classical controls versus the strongest released-setting QAOA row. QAOA time is the measured batch wall time divided equally among three methods.

Method	Trials	BKS hits	BKS rate	Near	Feasible	Seconds
HiGHS exact MILP	1	1	1.0000	1.0000	1.0000	0.036
Full-graph greedy	15,000	1,133	.07553	.31527	1.0000	19.95
Kernel greedy	15,000	6,063	.40420	.96873	1.0000	2.21
QAOA NL, Aer MPS	15,000	101	.00673	.15700	.9018	183.04

7 Audit trail and artifact map

The first validation and blind MPS pass used Qiskit transpilation at optimization level zero. Inspection of the released QOBLIB workflow showed that its Aer/MPS baseline receives native RZ/RZZ/RX circuits. Because basis decomposition changes where MPS truncation is applied, the first pass was preserved as a protocol audit and excluded from the main result. Champions were not reselected after this discovery; the corrected pass reevaluated the already frozen schedules. Training was exact-statevector simulation, where the gate representation does not change the unitary.

Table 8: Machine-readable artifact map. Paths are relative to the artifact root.

Purpose	Artifact
Frozen decisions	FROZEN_PROTOCOL.md
Deviation log	PROTOCOL_DEVIATIONS.md
Blind raw jobs	results/blind_test.json
Paired inference	results/analysis_summary.json
Sensitivity counts	results/mps_sensitivity.csv
Exact states and metrics	results/exact_mps_calibration.json
Compact exact table	results/exact_mps_calibration.csv
Classical controls	results/classical_baselines.json
Extended comparison	results/extended_comparison.json
cuTensorNet sweep	results/cutensornet_sweep.csv
Independent exact check	results/cutensornet/small_exact_validation.json
Independent attempt log	results/cutensornet/ATTEMPTS.md
Transpiled audit	results/blind_test_transpiled_audit.json
Integrity tests	test_full_cycle.py
Checksums and provenance	artifact_manifest.json

Eight integrity tests check the published ramp formula, split disjointness, the 186-to-55 reduction, disabled repair, balanced 15,000-shot blind artifacts, the 56-row exact-calibration matrix, tight-threshold convergence, and bond invariance above 16. They also check the completeness and validity of the classical controls and the independent exact/sweep artifacts. The analysis scripts regenerate the CSV tables, figures, statistical summaries, and reports from the JSON artifacts.